

Full LibriSpeech Copy-Move Forgery Dataset: A Large-Scale Audio Forensics Benchmark with Speaker-Disjoint Splits

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Abstract. Ensuring the authenticity of digital audio content has become critical as advanced editing tools grow more accessible. Copy-move forgery detection is particularly challenging as duplicated segments often evade traditional verification methods. Existing benchmarks suffer from limited size, speaker identity leakage, and inconsistent evaluation protocols. This paper introduces the Full LibriSpeech Copy-Move Forgery Dataset comprising 57,078 audio samples with speaker-disjoint splits, three forgery strengths, and detailed temporal annotations. The dataset’s scale (10× larger than previous benchmarks) and rigorous speaker separation enable robust model training and unbiased evaluation, addressing critical gaps that have limited deep learning approaches in audio forensics. Code and dataset: <https://github.com/RDisCoding/Full-LibriSpeech-CopyMove-Forgery-Dataset>

Keywords: Audio Forensics, Copy-Move Detection, Speaker-Disjoint Dataset, LibriSpeech, Deep Learning

1 Introduction

Recent years have witnessed a surge in the use and sophistication of audio editing technologies, making the manipulation of digital recordings both easier and more pervasive. Such capabilities have raised serious concerns in fields ranging from journalism to legal investigations, underscoring the vital need for reliable methods to authenticate audio evidence. One of the most challenging manipulation techniques is copy-move forgery, where a segment from the same recording is duplicated and repositioned, often in a manner that evades easy auditory detection due to preserved acoustic continuity and potential semantic changes [2, 3].

Progress in automated detection of such forgeries has been encumbered by significant limitations in available datasets [4, 7]. Prevailing benchmarks provide insufficient sample sizes—commonly under 5,000 samples—and are marred by speaker leakage between splits, reducing the reliability of system evaluations and inflating performance metrics [6]. Additionally, a lack of standardized evaluation protocols and diverse annotation schemes complicates fair comparison

across methods and impedes reproducible research. Without robust and scalable benchmarks, deep learning architectures are prone to overfitting and fail to generalize to real-world conditions.

Recent advances in the field have employed various techniques including spectral analysis methods [1, 5], machine learning approaches, and deep neural networks. However, the limited scale and diversity of available datasets continue to hinder progress in developing more robust detection methods that can generalize across different speakers, recording conditions, and forgery techniques. To address these limitations, we introduce the Full LibriSpeech Copy-Move Forgery Dataset, which makes the following contributions:

- i. A large-scale collection of 57,078 audio clips generated from the LibriSpeech train-clean-100 subset, offering a valuable resource for research in audio copy-move forgery detection that is $10\times$ larger than existing benchmarks, enabling robust deep learning model development.
- ii. Carefully designed speaker-disjoint splits for training, validation, and test sets to ensure that no speaker appears in more than one set, allowing unbiased and reliable evaluation.
- iii. Standardized audio clips of 10 seconds duration paired with high-resolution mel-spectrograms, making the dataset compatible with both classic signal processing methods and advanced deep learning models.
- iv. A set of forged samples categorized into three levels—subtle, medium, and obvious—enabling the evaluation of algorithms across varying degrees of manipulation complexity.
- v. Accurate annotations marking the exact locations of source and target segments within each forged sample, supporting the development of detection and localization techniques.
- vi. Consistent preprocessing steps and thorough documentation provided to facilitate reproducibility and fair comparison in forensic research.
- vii. Open access to the dataset, encouraging its adoption by the research community and promoting further advancements in automated audio forgery detection methods.

The remainder of this paper is organized as follows. Section 2 reviews related work in audio copy-move forgery detection and existing datasets. Section 3 details our dataset creation methodology. Section 4 presents the dataset characteristics and statistics. Section 5 discusses implications and future work, followed by conclusions in Section 6.

2 Related Work

Progress in audio copy-move forgery detection has depended heavily on the availability of high-quality benchmark datasets, yet existing resources remain limited in scale, diversity, and reproducibility. As shown in Table 1, most prior datasets contain fewer than 5,000 samples, creating substantial barriers to training and evaluating deep learning models. Such constraints reduce the data available for

Table 1. Comparison of Existing Audio Copy-Move Forgery Datasets. Reference numbers correspond to citations in the bibliography. The table demonstrates that our dataset significantly exceeds previous benchmarks in scale while ensuring speaker disjointness and public availability.

Dataset/Study	Orig. Files	Forg. Files	Atk. Spk. Files Disj.	Env. Info	Pub. Avail.
Xiao et al. [1]	-	-	-	-	No
Yan et al. [2]	1K	1K	3K	-	No
Huang et al. [3]	1K	1K	3K	-	No
Xie et al. [4]	1K	1K	1K	-	No
Wang et al. [5]	100	100	300	-	No
Su et al. [6]	1K	1.8K	-	-	No
Ustubioglu et al. [7]	158	398	2K	-	Yes
KTUCeng	600	1K	5.2K	Yes	Yes
Our Dataset	28.5K	28.5K	-	Yes	Clean

model development and weaken the reliability of generalization in forensic applications.

A major concern in many benchmarks is speaker leakage, where the same speaker appears in both training and testing splits. This issue inflates performance metrics by allowing models to learn speaker-specific traits instead of genuine forgery indicators. The absence of strict speaker separation compromises the credibility of reported results. In addition, many datasets lack standardized preprocessing pipelines and evaluation protocols, making fair comparison difficult. Variations in annotation quality, augmentation techniques, or metric definitions further hinder reproducibility across studies.

Existing benchmarks also suffer from limited diversity in speakers, environments, and languages, restricting model applicability beyond controlled settings. Temporal annotations are often sparse or imprecise, with most datasets lacking accurate localization ground truth for forged regions—an essential requirement for evaluating region-based detection methods.

Recent datasets such as KTUCengAudioForgerySet and LibriSpeech-derived resources attempt to address some of these issues by improving scale, metadata richness, and split integrity through speaker-disjoint partitions and additional annotations. However, even improved datasets—for instance, KTUCengAudioForgerySet with only 5,235 forged files—remain limited in size and comprehensiveness. Most still lack detailed temporal labels and standardized data formats needed for reliable benchmarking.

To address these gaps, the proposed dataset significantly expands sample size, enforces rigorous speaker-disjoint partitioning, introduces multi-level forgery difficulty, and provides high-resolution mel-spectrograms with fine-grained temporal annotations. Together, these features establish a stronger foundation for eval-

uating audio copy-move forgery detection and support the development of more robust, generalizable systems.

Table 1 highlights this advantage: our dataset contains 57,078 total files, far exceeding the largest existing dataset of 7,884 files. It is also the first large-scale benchmark to explicitly guarantee speaker-disjoint splits, ensuring that evaluation metrics reflect genuine generalization rather than speaker memorization.

3 Dataset Generation Methodology

Our dataset creation pipeline ensures data consistency, robustness, and applicability for audio forgery detection.

3.1 Source Data and Preprocessing

We use LibriSpeech’s train-clean-100 subset containing approximately 100 hours of clean English speech from 251 speakers. All audio files are loaded at 16 kHz sampling rate, converted to mono, and standardized to 10-second duration. Longer clips are randomly cropped and shorter ones zero-padded.

Speaker information is extracted from the LibriSpeech directory hierarchy. To avoid speaker-dependent bias, we create speaker-disjoint subsets using a fixed random seed (42): training (69.9%), validation (14.9%), and test (15.3%). This ensures models are evaluated on entirely unseen voices.

Quality control filters out files with over 80% silence or noise distortions, and excludes clips shorter than 5 seconds, ensuring only high-quality speech segments are retained.

3.2 Forgery Generation Process

For each original audio clip, we generated a corresponding forged version using our systematic copy-move algorithm. The forgery process follows these carefully designed steps to create realistic and challenging forgeries:

Strength Stratification. To simulate a range of manipulation scenarios, each forgery is categorized into three difficulty levels:

- i. *Subtle:* Segments with a duration of 0.3–0.8 seconds (average 0.55 s). These short insertions resemble quick, natural repetitions in speech and are therefore harder to detect.
- ii. *Medium:* Segments ranging between 0.5 and 1.5 seconds (average 1.0 s). These duplications are more noticeable than subtle ones but still moderate in length, striking a balance between naturalness and detectability.
- iii. *Obvious:* Segments spanning 1.0–2.5 seconds (1.75 s). These are long and clearly repeated portions, making them the easiest category to spot.

Segment Selection. For each clip, we randomly select a source segment within the first 60% of the audio to ensure sufficient space for target placement. The segment duration is chosen uniformly within the strength category range to avoid bias toward specific durations.

Target Placement. The copied segment is pasted at a target location that maintains the minimum gap constraint and fits within the audio duration. If no valid target location exists (due to duration constraints), we apply fallback strategies including slightly reducing segment duration or adjusting gap requirements to ensure successful forgery generation.

Temporal Annotation. Each manipulation is annotated with exact start and end times for both source and target regions, recorded with millisecond precision. These annotations serve as ground truth for evaluating both binary forgery detection (whether a clip is manipulated) and fine-grained localization (where manipulation occurs). The design of this annotation step links directly to the spectrogram generation stage, where such repetitions manifest as identifiable patterns. Figure 1 illustrates the forgery generation process with representative waveform examples showing how copy-move operations create characteristic patterns in both time and frequency domains.

3.3 Spectrogram Generation

To support image-based learning methods, we transform each audio file into a standardized mel-spectrogram. This conversion bridges the gap between raw audio and convolutional neural network (CNN) architectures commonly used in visual tasks. The spectrograms are generated with the following configuration: (1) **Mel bands:** 128, providing sufficient frequency resolution for speech signals; (2) **FFT window size:** 2048 samples (≈ 128 ms at 16 kHz), balancing temporal and frequency precision; (3) **Hop length:** 512 samples, yielding 75% overlap for smooth continuity; (4) **Dynamic scaling:** power values are converted to decibels and normalized to a common reference.

Each spectrogram is rendered as a PNG image using the *viridis* colormap and stored at 150 DPI resolution, yielding fixed-size images of 128×313 pixels. These spectrograms clearly highlight repetitive energy bursts caused by copy-move forgeries, making them suitable for CNN-based detection approaches. Figure 2 shows representative spectrograms for original and forged audio, clearly demonstrating how copy-move operations create visible repetitive patterns in the time-frequency representation that can be exploited by CNN-based detection methods.

4 Dataset Characteristics and Statistics

4.1 Dataset Composition

The dataset contains 57,078 audio files: training set with 39,874 files (69.9%), validation set with 8,478 files (14.9%), and test set with 8,726 files (15.3%). Each

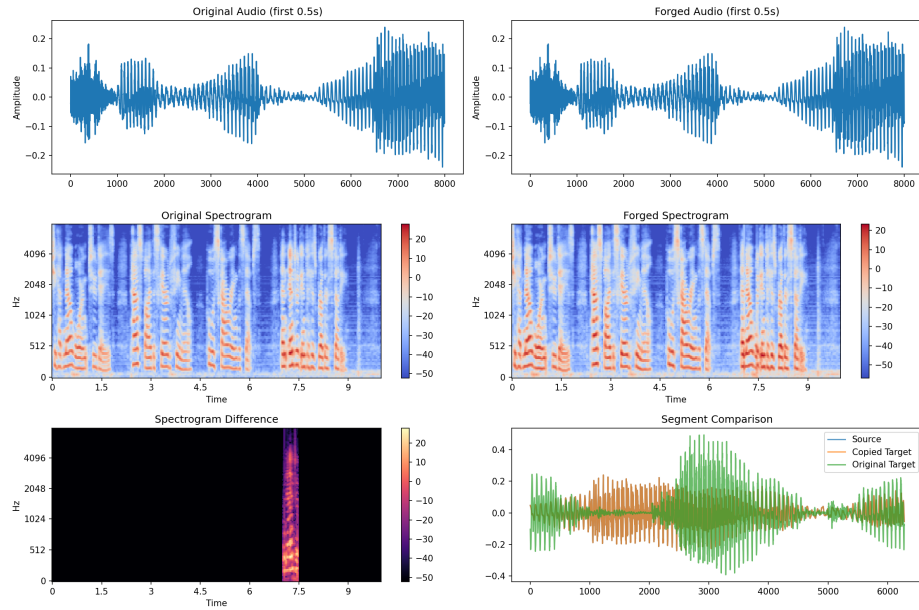


Fig. 1. Illustration of the copy-move forgery process showing comprehensive analysis across multiple representations. Top row shows waveform comparison between original (left) and forged (right) audio for the first 0.5 seconds, demonstrating identical temporal patterns. Middle row displays complete mel-spectrograms for original (left) and forged (right) 10-second clips showing full frequency-time evolution. Bottom row presents spectrogram difference analysis (left) highlighting regions where forgery occurred as bright spots, and segment comparison (right) showing temporal alignment between source segment (orange), copied target segment (orange), and original target location (green), demonstrating how copy-move operations create detectable artifacts in both time and frequency domains.

split maintains equal original and forged samples, preventing class imbalance. Speaker identities remain completely disjoint across splits. The 57,078-sample scale addresses a critical limitation in existing benchmarks: most prior datasets contain fewer than 5,000 samples, which is insufficient for training modern deep learning architectures that typically require tens of thousands of samples to learn robust, generalizable representations without overfitting.

4.2 Forgery Strength Distribution

Table 2 presents the distribution of forgery strengths across the dataset. We maintain an almost equal representation of subtle, medium, and obvious categories within each split. This balanced distribution ensures unbiased evaluation across the full spectrum of forgery difficulty, prevents models from overfitting to specific manipulation patterns, and enables systematic analysis of detection performance as a function of forgery complexity.

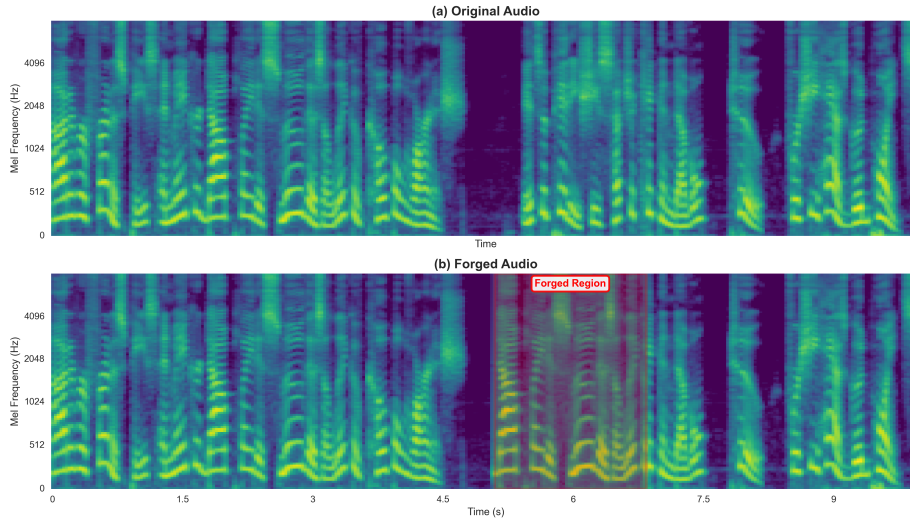


Fig. 2. Example mel-spectrograms demonstrating copy-move forgery patterns in frequency domain. (a) Original audio spectrogram showing natural speech spectral evolution with formant transitions and phonetic variations typical of continuous speech, (b) Forged audio spectrogram showing duplicated spectral patterns (highlighted regions) where the same frequency-time patterns appear at different temporal locations, creating characteristic repetitive structures. The duplicated regions show identical formant patterns, harmonic structures, and spectral envelopes that distinguish forged content from natural speech variations.

4.3 Temporal Characteristics

The forged segments in the dataset exhibit diverse temporal properties that differentiate forgery strengths while maintaining intra-class variability. Segment durations span from 0.3 to 2.5 seconds, with an overall mean of 1.1 seconds and a standard deviation of 0.62 seconds. Gap distances between source and inserted segments range from 0.2 to 9.3 seconds, with a mean of 2.8 seconds and a standard deviation of 2.1 seconds, ensuring sufficient separation to prevent trivial detection based on temporal proximity. Figure 3 illustrates the temporal statistics for each strength category. The segment duration distribution reveals distinct ranges for subtle (0.3–0.8 s), medium (0.5–1.5 s), and obvious (1.0–2.5 s) categories, with controlled overlaps designed to create graduated difficulty. The gap distributions demonstrate an exponential decay trend with a mean of 2.8 seconds, ensuring realistic audio structure and establishing duration, rather than temporal gap, as the primary factor influencing detection difficulty.

4.4 Speaker Demographics

The dataset incorporates speakers with diverse demographic characteristics derived from the original LibriSpeech corpus. It contains 57,078 speaker–file com-

Table 2. Distribution of Forgery Strengths Across Dataset Splits. The balanced representation (variation $<0.03\%$) across categories prevents artificial performance bias and ensures fair evaluation of detection methods across all forgery difficulty levels.

Split	Subtle	Medium	Obvious	Total Forged
Train	6,646	6,645	6,646	19,937
Validation	1,413	1,413	1,413	4,239
Test	1,453	1,454	1,456	4,363
Total	9,512	9,512	9,515	28,539

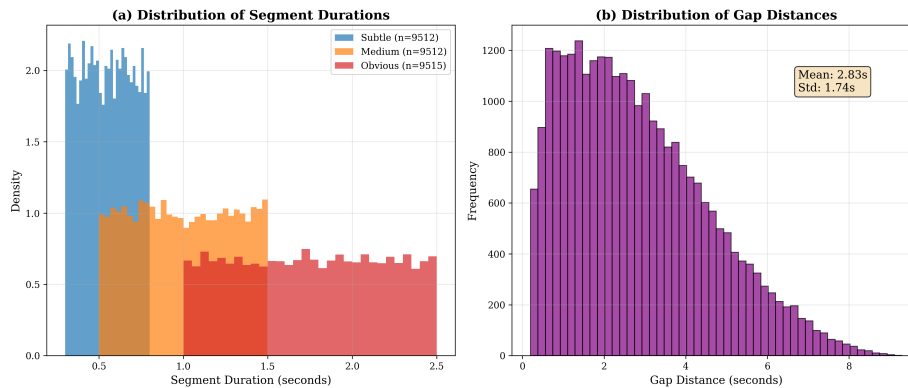


Fig. 3. Temporal characteristics of forged segments across strength categories. (a) Distribution of segment durations showing clear separation between subtle (0.3–0.8s), medium (0.5–1.5s), and obvious (1.0–2.5s) categories with controlled overlaps for graduated difficulty. (b) Distribution of gap distances between source and target segments following an exponential decay with mean gap of 2.8s, ensuring sufficient temporal separation while maintaining realistic audio structure.

binations from 251 distinct speakers, with each speaker contributing an average of 227 files (median: 211, ranging from 40 to 510). This distribution supports adequate diversity while providing a sufficient number of samples per speaker for robust statistical analysis.

The gender distribution reflects metadata availability in LibriSpeech: 8.6% female speakers (4,888 files), 9.2% male speakers (5,260 files), and 82.2% speakers with unspecified gender information (46,930 files). Although gender information is incomplete, the dataset maintains broad speaker diversity to ensure model robustness across varied vocal characteristics. Figure 4 illustrates speaker distribution patterns across dataset splits and demographics, confirming careful partitioning that upholds evaluation integrity.

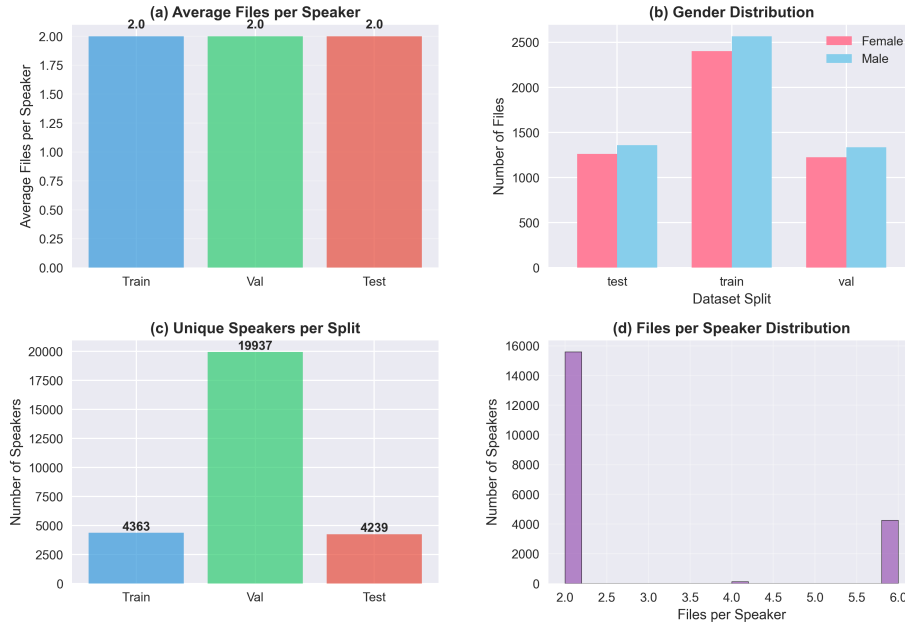


Fig. 4. Speaker distribution analysis across dataset splits ensuring evaluation integrity. (a) Number of files per speaker in each split showing balanced contribution across speakers with consistent distribution patterns, preventing dominance by individual speakers. (b) Gender distribution across splits showing proportional representation of available metadata (8.6% female, 9.2% male, 82.2% unknown). Consistent proportions across splits ensure that acoustic gender differences do not confound evaluations, while the large ‘unknown’ group mirrors limitations of the original LibriSpeech metadata.

5 Limitations and Future Improvements

While the proposed dataset marks a significant advance for audio forgery detection research, several limitations remain for future exploration. First, the current release focuses solely on copy-move forgeries; incorporating splicing, deepfake synthesis, and hybrid manipulations would offer a more comprehensive benchmark for general-purpose detection models. Second, real-world forgeries often undergo post-processing operations such as noise addition, compression, and filtering, which can mask forensic cues. Including such variations would enable systematic evaluation of model robustness under realistic conditions. Third, cross-dataset testing remains largely unexplored and would provide deeper insights into model generalization across diverse domains. Finally, the dataset’s restriction to English speech limits its applicability in multilingual settings; expanding to additional languages and dialects would enhance its global relevance and practical utility.

6 Conclusion

This work introduces the Full LibriSpeech Copy-Move Forgery Dataset, a large-scale benchmark addressing key limitations in audio forensics research. With 57,078 audio files, speaker-disjoint partitions, and comprehensive temporal annotations, it provides a reliable foundation for developing and evaluating forgery detection methods. Key contributions include unprecedented scale ($10\times$ larger than previous benchmarks), strict speaker disjointness, three-tier strength categorization, and standardized preprocessing ensuring reproducibility. By making this resource publicly available, we accelerate research and establish a community benchmark. Future work will integrate additional attack types and post-processing variations to advance comprehensive audio authentication tools.

Data Availability. The dataset is available upon request, subject to completion of a request form. The baseline implementation code has been open-sourced and can be accessed at: <https://github.com/RDisCoding/Full-LibriSpeech-CopyMove-Forgery-Dataset>

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